1D filter $h = \frac{1}{4}[1, 2, 1]$ (k = -1, 0, 1)

Signal $y[t] = \cos(\omega t)$

convolution (∞ but $h(k,t)$ support is finite in k (in this example, $h \neq 0$ iff $k \in [-1, 1]$))

$$(h * y)[t] = \sum_{k=-\infty}^{\infty} h[-k]\, y[t+k]$$

(conv vs corr)

$$= h[1]\, y[t-1] + h[0]\, y[t] + h[-1]\, y[t+1]$$

$$= \frac{1}{4}\cos(\omega(t-1)) + \frac{1}{2}\cos(\omega(t)) + \frac{1}{4}\cos(\omega(t+1))$$

$$= \frac{1}{4}\,\mathrm{Re}\left[e^{i\omega t}\left(\underbrace{e^{i\omega} + e^{-i\omega}}_{2\cos(\omega)}\right)\right] + \frac{1}{2}\cos(\omega t)$$

$$= \frac{1}{2}\cos(\omega)\cos(\omega t) + \frac{1}{2}\cos(\omega t)$$

$$= \underbrace{\frac{\cos(\omega) + 1}{2}}_{\text{frequency-dependent constant}} \cos(\omega t)$$

same frequency as source

gain/attenuation based on frequency.

low-pass filter => keeps low freq, attenuate high freqs.

high-pass => vice versa